

Experiment 13

Handle missing values and inappropriate data in a given data set.

Program:

```
# Upload Hotel.csv and convert it into DataFrame
import numpy as np
import pandas as pd

df=pd.read_csv("Hotel_Dataset.csv")
df

# Identify duplicate rows
df.duplicated()
df.info()

# Remove duplicate rows and reset the index
df.drop_duplicates(inplace=True)
len(df)

index=np.array(list(range(0, len(df))))
df.set_index(index,inplace=True)

# Remove unwanted column
df.drop(['Age_Group.1'],axis=1,inplace=True)

# Replace inappropriate numerical values with NaN
df.CustomerID.loc[df.CustomerID<0]=np.nan
df.Bill.loc[df.Bill<0]=np.nan
df.EstimatedSalary.loc[df.EstimatedSalary<0]=np.nan
df['NoOfPax'].loc[(df['NoOfPax']<1) | (df['NoOfPax']>20)]=np.nan

# Correct inconsistent categorical values
df.Hotel.replace(['Ibys'],'Ibis',inplace=True)
df.FoodPreference.replace(['Vegetarian','veg'],'Veg',inplace=True)
df.FoodPreference.replace(['non-Veg'],'Non-Veg',inplace=True)

# Fill missing values
df.EstimatedSalary.fillna(round(df.EstimatedSalary.mean()),inplace=True)
df.NoOfPax.fillna(round(df.NoOfPax.median()),inplace=True)
df['Rating(1-5)'].fillna(round(df['Rating(1-5)'].median()),inplace=True)
df.Bill.fillna(round(df.Bill.mean()),inplace=True)

df
```

Output

Output:
Duplicate rows are identified and removed.
The duplicate record reduces the dataset to 10 rows.
Invalid negative values are converted to NaN.
Invalid NoOfPax values are converted to NaN.
" Ibys " is corrected to "Ibis".
"Vegetarian" and "veg" are corrected to "Veg".
"non-Veg" is corrected to "Non-Veg".
Missing continuous values are filled using the mean.
Missing discrete values are filled using the median.
Missing categorical values are handled using the mode.